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# MILITARY SPECIFICATION

# SAW, ABRASIVE DISK, CONCRETE: ENGINE-DRIVEN, SELF-PROPELLED, 12- TO 24-INCH BLADE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers an engine-driven, self-propelled saw using a circular blade for sawing concrete and asphalt materials containing hard aggregate.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

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| QQ-S-781 | - Strapping, Steel and Seals. |
| TT-P-664 | - Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |

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| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-B-121 | - Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-C-5501 | - Caps and Plugs, Protective, Dust and Moisture Seal. |
| MIL-E-10062 | - Engines: Preparation for Shipment and Storage of. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-L-21260 | - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In. |
| MIL-T-22085 | - Tape, Pressure-Sensitive, Adhesive Preservation and Sealing. |
| MIL-C-46168 | - Coating, Aliphatic Polyurethane, Chemical Agent Resistant. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-595 | - Colors. |
| FED-STD-H28 | - Screw-Thread Standards for Federal Service. |

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| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-209 | - Slinging Eyes and Attachments for Lifting and Tying Down of Military Equipment. |
| MIL-STD-838 | - Lubrication of Military Equipment. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods. |



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| MIL-STD-1400 | - Engine, Gasoline or Diesel, Methods of Test. |
| MIL-STD-1472 | - Human Engineering Design Criteria for Military Systems, Equipment and Facilities. |
| MIL-STD-1473 | - Standard General Requirements for Color and Marking of Army Material. |
| MIL-STD-1474 | - Noise Limits for Army Material. |

HANDBOOK

MILITARY

| | |
|---|---|
| MIL-HDBK-113 | - Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems. |

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM A 108 | - Specifications for Cold-Finished Carbon Steel Bars and Shafting. |
| ASTM A 331 | - Specifications for Cold-Finished Alloy Steel Bars. |
| ASTM A 366 | - Specifications for Cold-Rolled Carbon Steel Sheets and Strip. |
| ASTM A 569 | - Specifications for Hot-Rolled Carbon Steel Sheets and Strip, Commercial Quality. |
| ASTM D 3951 | - Standard Practice for Commercial Packaging. |

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

RUBBER MANUFACTURER'S ASSOCIATION (RMA)

RMA 1P-20-1977 - Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturer's Association, 444 Madison Avenue, New York, NY 10022.)



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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J88 - Sound Measurement - Earthmoving Machinery - Exterior.
J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Description. The saw shall be self-propelled, chassis mounted and engine-driven with guide, water-pump, and the necessary controls and guards. The saw shaft shall be adaptable to cut on either side of the chassis with 12-, 14-, 18- and 24-inch diameter saw blades from 1/8- to 3/8-inch wide.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the saw from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish the saw conforming to this specification.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The saw shall be fabricated from compatible materials that are inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the saw may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with



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components and subcomponents, and shall make information available upon request to the contracting officer or the designated representative (see 6.2).

3.3.4 Common parts. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes and shall be threaded in accordance with FED-STD-H28.

3.3.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the saw may be newly fabricated from recovered materials to the maximum extent practicable, provided the saw produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incoporated in the saw.

3.3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.4 Safety.

3.4.1 Safety. All exposed parts which are subject to high operating temperature or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or located such that they are a hazard to operating personnel shall be fully enclosed or guarded. Safety guards shall be mounted and shall either be removable or shall be provided with access doors or openings for servicing and inspection. Protective devices shall not impair the operating and maintenance functions. The engine exhaust gases shall be directed away from the operator. Steel guards shall be provided for the power transmission belts and saw blade. The saw guard shall be constructed such that the guard may be used on both sides of the saw. The trailing edge of the blade guard shall be fitted with a rubber shield that will contact the surface when the saw blade contacts the surface.

3.4.2 Noise limits. The noise produced by the saw shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.2.13. The provisions of MIL-STD-1474, 4.3, and 4.4, shall be provided if and only if the procedures outlined in 5.1.1.2 of MIL-STD-1474 have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with 4.3 of MIL-STD-1474, and be readable at the distance indicated on the sign.

3.4.3 Exterior noise. The saw shall meet the requirements of MIL-STD-1474 for exterior noise construction and materials-handling equipment when tested in accordance 4.5.2.14.

3.5 Human factors engineering design requirements. The saw shall conform to accepted human factors engineering design criteria, as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls)



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5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design for Maintainability), 5.13 (Hazards and Safety), of MIL-STD-1472, as applicable.

3.6 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to the gear case. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.5.2.1 shall be accomplished by not more than two people (except a hoist and operator may be used as necessary) using only the special tools furnished with the saw and common tools. After completion of any maintenance operation, the saw shall operate as specified herein.

3.7 Engine and accessories.

3.7.1 Engine. The saw shall be powered by a gasoline or diesel engine as specified (see 6.2) and shall be an air-cooled, commercial industrial type, rated at not less than 30 horsepower. The saw under any of the requirements and test specified herein shall not require horsepower in excess of the continuous horsepower rating of the engine at governed speed. The engine shall produce the required performance on unleaded or low-lead content gasoline if it is a gasoline type engine, or on diesel fuel if it is a diesel type engine (see 6.2). The engine shall be equipped with a fuel tank with a capacity of not less than 5 gallons, a housing with removable side panels, electrical 12 volt cranking system, a battery charging generator or alternator system of not less than 25 ampere capacity and an instrument panel equipped with the following:

a. Engine lubricating oil pressure gage.
b. Battery charging ammeter or volt meter.
c. Mechanical or electrical hour meter.

The instrument panel shall be located in full view of the operator during all normal operating conditions.

3.7.2 Batteries. Batteries shall be manufacturer's standard commercial battery. Batteries shall be shipped charged and dry. The batteries shall be protected from the weather by a storage compartment which has drain holes and is treated with acid resistant paint. The battery terminals shall be provided with guards to prevent accidental shorting.

3.8 Blade shaft and drives.

3.8.1 Blade shaft. The blade shaft shall be fabricated of solid steel conforming to ASTM A 108 or A 331, grade optional, and shall be mounted on double tapered roller bearings. A 1-inch diameter arbor shall be provided on each end of the blade shaft. Provisions shall be made to prevent the blade shaft from turning while removing and replacing the blade. The shaft must allow the saw blade



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to be attached to either side. The diameter of the collars shall be such that, when used with the respective saw blades, the following minimum cutting depths shall be possible:

| Saw blade diameter (inches) | Cutting depth (inches) |
|---|---|
| 12 | 3-1/2 |
| 14 | 4-1/2 |
| 18 | 6-1/2 |
| 24 | 9 |

3.8.2 Drives. The engine torque shall be transmitted to the blade shaft by V-belts. The V-belt drives and the V-belts shall conform to RMA 1P-20-1977. The V-belt tension shall be adjustable. The saw shall be furnished equipped for operation with saw blades either 12 or 14 inches in diameter; in addition, the necessary parts and components to adapt the saw for operation with blades 18 and 24 inches in diameter shall be furnished.

3.9 Saw blades. When specified (see 6.2), saw blades shall be furnished in the following types, sizes, and quantities:

| Quantity | Type and size of blade |
|---|---|
| 4 | 12-inch diamond blades |
| 4 | 14-inch abrasive blades (wet and dry type) |
| 2 | 14-inch diamond blades |
| 2 | 18-inch abrasive blades (wet and dry type) |
| 2 | 24-inch diamond blades |

3.10 Box for accessories, blades, and tools. A separate weatherproof box fabricated of steel conforming to ASTM A 366 or A 569, 0.0747-inch in nominal thickness, shall be furnished with each saw. The box shall have the capacity to store the drive accessories specified in 3.8.1 and 3.8.2, the blades specified in 3.9, and the tools specified in 3.6. The box shall have built-in restraints such as compartments or holddowns to keep each item or group of items in their respective place regardless of the position in which the box is placed. The lid of the box shall open at least 90 degrees and shall be mounted on continuous (piano type) hinges. Two tension, hook-type fasteners shall be provided that will keep the lid closed when the toolbox is subjected to vibration. A hasp that will accommodate a padlock with a 3/8-inch shackle shall be provided for locking the box. The box shall be equipped with handles.

3.11 Saw pointer guide. The saw shall be furnished with two adjustable cutting-line pointers. One pointer shall be located in front of the saw and one shall be located to the rear of the saw. The pointers shall be mountable on either side of the saw.

3.12 Depth control. The saw shall be furnished with hydraulic-type control to adjust the cutting depth of the saw. The control shall have a locking device



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to maintain any cutting depth setting. The control shall have an adjustment to raise a 24-inch saw blade clear of the surface and to lower a 12-inch blade 3-1/2-inches into the cut. A gage with graduations no more than 1/2 inch shall be furnished to indicate the depth at which the saw is cutting. The depth of cut shall be within plus or minus 1/4-inch of the depth indicated on the depth gage. The gage shall have graduations for 12- to 24-inch saw blades.

3.13 Water system. The saw shall be equipped with a complete blade coolant system and a low water pressure switch. The necessary valves, controls, piping and hose to spray water on both sides of the saw blade, whether mounted on right or left side, shall be furnished. Provisions shall be made to attach a 3/4-inch garden hose to the water system. If required, a standard 3/4-inch garden hose adapter shall be provided. A means of completely draining the water system to prevent damage from freezing shall be provided.

3.14 Propelling mechanism. The saw shall be propelled by the engine through a hydrostatic transmission. Power shall be transmitted from the engine to the transmission by V-belts conforming to 3.8.2 and from the transmission to the drive wheels by power transmission chains or by gears. The travel speed shall be adjustable to any speed between 0 and 150 feet per minute.

3.15 Wheels. The saw shall be furnished with four non-castering wheels mounted on antifriction bearings. The wheels shall be solid rubber, or cast iron with solid rubber, and shall be arranged with the axles to provide a rigid suspension. The driving forces shall be transmitted to the driving wheels through a point on the wheels that will not contact the surface of the roadway.

3.16 Parking brake. The saw shall be furnished with a brake or a similar mechanism to prevent the saw from free rolling when the saw is placed on a 10 percent grade with the longitudinal axis of the saw parallel to the grade.

3.17 Control and instrument panel. Handles for one-person operation shall be provided at the rear of the saw. All hand-operated controls and instruments shall be mounted on a panel accessible and visible to the operator.

3.18 Lubrication. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.18.1 Lubrication fittings. Lubrication fittings shall be the manufacturer's standard as furnished commercially. Fittings shall be located in a protected position and shall be accessible to a lever-operated, hand-type, grease gun. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.18.2 Pressure-release device. An automatic pressure-release device shall be provided where pressure lubricating equipment will damage grease seals or other parts.



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3.18.3 Filling, draining, and checking provisions. Enclosures, such as gear cases and transmission housing, that contain a reservoir of lubricants shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling with lubricant and for draining without draining onto the saw. The drain plug shall be of the magnetic type and located so that removal of the plug will result in complete drainage of the lubricant from the enclosure.

3.18.4 Grease lubrication. All grease lubrication shall be provided in accordance with MIL-G-10924. The saw shall be assembled, run-in, tested, and delivered with these greases. Each component shall be tagged in a conspicuous place to indicate what grease has been used.

3.18.5 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.19 Lifting attachments. Permanently affixed attachments that enable the saw to be lifted in its normal travel positions shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209 for type IV equipment, class 1 and 3 provisions. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.20.

3.20 Tiedown attachments. Permanently affixed attachments that permit fastening the saw to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209 for type IV equipment, class 2 or 3 provisions. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.19. All lift/tiedown points shall be labeled "Lift", "Tiedown", or "Lift-Tiedown", as appropriate, in one-inch high letters.

3.21 Identification marking/labels. The saw shall be identified in accordance with MIL-STD-130. All marking/labels shall be applied to the saw on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.22 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition, shall show the silhouette of the saw in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.23 Control identification, safety, warning, and information plates. Unless otherwise specified (see 6.2) a plate shall be furnished for each switch, gage, and control and at points that require special attention for safe and efficient operation of the saw. The plates shall be labeled by lettering or other appropriate marking with information such as function, direction the control is to be moved, neutral position, safety instruction, and other warnings as necessary.



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A plate labeled with the recommended engine speed for each size and type of blade shall be mounted in a conspicuous location. The plates shall conform to MIL-P-514, type II, composition C, grade A, class 1 and shall be not less than 0.028-inch thick. Safety and warning plates shall be in accordance with MIL-STD-1473. Plates that describe safety measures or that warn against operations detrimental to the saw shall be distinguished by the words "CAUTION" or "WARNING" in red lettering matching color number 31136 of FED-STD-595.

3.24 Treatment and painting. The portions of the saw normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) the top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.25 Stenciling. The gross weight of each saw shall be stenciled on each side of the saw so as to be discernible. All stenciling shall be in block type letters not less than 1-inch high, and shall be black in color.

3.26 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptibility characteristics of the saw shall conform to SAE J551, devices category.

3.27 Workmanship. The saw shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the saw. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.28 Items to be furnished with the saw. No shipment of saws shall be made unless repair parts, maintenance tools, and accessories are included with the shipment, unless approval for shipment of the saw without such items has been received from the contracting officer (see 6.2).

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a



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part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection. Prior to examination and test of the saw, the engine which is to be installed in the first article model shall be tested in accordance with test method series 2000, MIL-STD-1400. Nonconformance to 3.7.1, or failure to meet the requirements of test method 2000 of MIL-STD-1400 or the operational test specified herein shall constitute failure of this test.

4.3.1 Examination. The saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The saw shall be tested as specified in 4.5.2.1 through 4.5.2.15. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each saw shall be tested as specified in 4.5.2.9. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Examine the saw for the following defects:

101. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
102. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
103. Contractor does not have documentation available for identification of material, material finishes or treatment.
104. Common parts not as specified.
105. Material not suitable or of incorrect thickness.
106. Safety and human factor elements do not meet requirements as specified.



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107. Engine not as specified.
108. Batteries not as specified.
109. Blade shaft not as specified.
110. Drives not as specified.
111. Saw blades not as specified.
112. Box for accessories, blades, and tools not as specified.
113. Saw pointer guide not as specified.
114. Depth control not as specified.
115. Water system not as specified.
116. Propelling mechanism not as specified.
117. Wheels not as specified.
118. Parking brake not as specified.
119. Control and instrument panel not as specified.
120. Lubrication fittings not as specified.
121. Pressure-release device not as specified.
122. Filling, draining, and checking provisions not as specified.
123. Grease lubrication not as specified.
124. Oil lubrication not as specified.
125. Identification marking and labels not as specified.
126. Control identification, safety, warning and information plates not as specified.
127. Treatment and painting not as specified.
128. Color not as specified.
129. Stenciling not as specified.
130. Electromagnetic interference not as specified.
131. Workmanship not as specified.
132. Saw does not conform to 3.5 human factors engineering.
133. Lubricants not as specified.
134. Lifting provisions not as specified.
135. Tiedown provisions not as specified.
136. Shipping data plate not as specified.
137. Used, rebuilt, or remanufactured components, pieces, and parts incorporated within the saw.
138. Interchangeability not as specified.
139. Threaded parts not in accordance with FED-STD-H28.
140. Repair parts, maintenance tools, and accessories not included with shipment of saws.

4.5.2 Test.

4.5.2.1 Service and maintenance. Perform the following service and maintenance operations:

a. Check and adjust the battery electrolyte level.
b. Check engine crankcase and all other gear cases and refill each to operating level.
c. Lubricate all points which require grease lubrication.
d. Service engine air cleaner.
e. Evaluation shall be accomplished throughout testing to determine compliance with 3.4 and 3.5.



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Inability to perform any of the service and maintenance operations, or safety features missing or not as specified shall constitute failure of this test.

4.5.2.2 Operational. Concrete made of portland cement and hard aggregate and cured for not less than 30 days shall be used for the test. Mount a 12-inch, diamond-type blade on the right side of the saw (see 3.8.1). Operate the saw to cut the concrete to a depth of 3-1/2-inches in increments of 1/2-inch as determined from the depth gage (see 3.12). When the saw has cut to the depth for which the gage is set, the cut shall be extended forward not less than 1 foot with the propelling mechanism propelling the saw into the cut. This procedure shall be repeated for each 1/2-inch incremental depth setting, measuring the depth, and observing the depth of cut indicated on the depth gage. Repeat the test with the blade mounted on the left side of the saw. One or more of the following shall constitute failure of this test:

a. A total cutting depth of less than 3-1/4 inches.
b. Depth of cut not within plus or minus 1/4-inch of the depth indicated on the depth gage for each incremental cut.
c. Inability of the saw to propel itself into the cut.
d. Inability to operate with the blade on the right or left side of the saw.

4.5.2.3 24-inch blade. Mount a 24-inch blade using the parts and components specified in 3.8.1 to adapt the saw for operation. Operate the saw to cut the concrete to a depth of 8-1/2-inches. Either of the following shall constitute failure of this test:

a. Inability to convert the saw for use with a 24-inch blade from parts and components furnished.
b. A total cutting depth of less than 8-1/2-inches.
c. Insufficient engine horsepower.

4.5.2.4 Depth control. During the test specified in 4.5.2.2 and 4.5.2.3, the depth control shall be checked (see 3.12). One or more of the following shall constitute failure of this test:

a. Inability of the depth control to raise and lower blades to the extent specified.
b. Inability of the depth control to maintain setting.

4.5.2.5 Water system. During the test specified in 4.5.2.2 and 4.5.2.3 the blade cooling system shall be checked. During the test decrease the water pressure to activate the low water pressure switch (see 3.13). One or more of the following shall constitute failure of this test.

a. Water leaks.
b. Inability to cool the blade.
c. Low water pressure switch fails to stop the engine.



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4.5.2.6 Travel speed. Raise the blade above the surface. Operate the saw with the propelling mechanism. Vary the traveling speed from 0 to 150 feet per minute (see 3.14). Determine the travel speed of the saw. Either of the following shall constitute failure of this test:

a. Inability to vary the traveling speed of the saw to any speed between 0 and 150 feet per minute.
b. Maximum forward speed less than 150 feet per minute.

4.5.2.7 Brake. Place the saw on a 10 percent grade with the longitudial axis of the saw parallel to the grade. Raise the saw blade above the surface. Apply the parking brake or similar mechanism (see 3.16). Evidence of the saw free rolling shall constitute failure of this test.

4.5.2.8 Electromagnetic interference. Unless otherwise specified, the saw shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article model test report (see 3.26).

4.5.2.9 Individual. Operate each saw and determine the functioning of all components and controls (see 3.17). Faulty operation of any component or control or leakage of lubricants shall be cause for rejection.

4.5.2.10 Engine power. The engine power test shall be performed while the saw is cutting concrete as specified in 4.5.2.3. The power required to operate the saw shall be measured by an indirect method utilizing data relating intake mainfold pressure versus horsepower. When the indirect method is utilized, the minimum and maximum engine rpm and intake manifold vacuum shall be recorded while operating under the above conditions. These readings shall be compared with the engine speed (rpm), brake horsepower and vacuum curve, for the continuous brake horsepower as established during conduct of test method series 2000, of MIL-STD-1400 (see 3.7.1). Any intake manifold vacuum reading recorded during operation of the saw that is lower than the reading listed on the established horsepower curve at the corresponding speed (rpm) shall constitute failure of this test.

4.5.2.11 Lifting provisions.

a. Lift the saw and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the saw. Determine the forward angles of application for each provision. Measure the clearance between each sling and the saw. Measure eye opening and clearance dimensions of the provision. Measure the load carried by each provision.
b. Restrain the saw by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209, for type IV equipment, class 1 and 3 provisions times the load it carried in "a" above. An alternate method of test is to lift the saw as in "a" above and add weights to the main frame until the required force is obtained. Hold the saw under either of the above loading conditions for at least 90 seconds.



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4.5.2.11.1 Failure criteria. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, for type IV equipment, class 1 and 3 provisions as specified in 3.19 shall constitute failure of this test.

4.5.2.12 Tiedown provisions. With the saw anchored by means other than the attachment being tested, apply the proportion of the static loads as specified in MIL-STD-209, for type IV equipment, class 2 and 3 provisions to each attachment. Measure dimensions of the tiedowns.

4.5.2.12.1 Failure criteria. Failure to meet the requirements of MIL-STD-209, for type IV equipment, class 2 or 3 provisions as specified in 3.2 shall constitute failure of this test.

4.5.2.13 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position and, additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.14 Exterior spectator. Measure the exterior noise level in accordance 4.5.2.14 with SAE J88. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.2.15 Performance testing. This test shall be an operational performance test for a period of not less than 8 hours. The test shall consist of sawing concrete made of portland cement with a compressive strength of not less than 2000 psi while traveling. The travel time using the saw's own propelling mechanism shall not be greater than 20 percent of the test hours specified herein. The concrete sawing shall be accomplished by using the blades and related cutting depths specified in 3.8 with an equal amount of sawing time for each size blade specified until the specified test time is complete. Failure to complete the test, inability to operate and function as specified, malfunction of any control, leakage of fuel, lubricants, or water shall constitute failure of this test.

4.6 Inspection comparison. The Government may select saws at any time during the contract production period and subject these saws to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article saws. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.



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4.6.1 Inspection failure. Failure of any saw to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the saw and shall be cause for refusal by the Government to continue acceptance of production saws until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on saws previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all saws accepted after approval of the first article saw are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. Examine the first article pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Tests. When specified (see 6.2), the first article pack of a boxed saw for level A or B shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the saw shall constitute failure of the first article pack and shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completely processed saw prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

141. Disassembly not as specified for level A or B.
142. Unpainted exterior unfinished metal surfaces not coated with preservative for level A or B.
143. Exposed exterior machined surfaces not coated with preservative, wrapped or covered with barrier material, and barrier material not secured as speecified for level A or B.
144. Engine and engine accessories not preserved in accordance with the referenced document for level A, B, or C.
145. Drive belts not removed or released from tension for level A or B.
146. Unpainted pulley grooves not coated with primer for level A.
147. Gears not preserved as specified for level A or B.



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148. Drive chains not preserved as specified for level A or B.
149. Components of the depth control not preserved as specified for level A or B.
150. Water pump not preserved and shutoff cock not opened as specified for level A or B.
151. Repair parts not preserved in accordance with the referenced documents for level A or B.
152. Maintenance tools not preserved in accordance with the referenced document for level A or B.
153. Consolidation not as specified for level A or B.
154. Packing not as specified for level A or B.
155. Strapping not as specified for level A or B.
156. Depreservation guide not prepared, placed in a waterproof envelope, and secured on the saw.
157. Marking missing, illegible, incorrect, or incomplete.

## 5. PACKAGING

5.1 First article. When specified (see 6.2), the contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging, that the applied preservation, packing, and marking comply with the packing requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The first article pack may be accomplished utilizing either the first article model saw or a production saw. If the first article model saw is utilized, any preservation and packing test shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article model saw and production saws.

5.2 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly of the saws shall be confined to the removal of protruding parts that would otherwise increase cubage in accordance with the contractor's recommendations. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss. Removed keys shall be secured to the keyway of the primary component or shaft.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected surfaces.

5.2.1.3.1 Unfinished surfaces. Unpainted exterior unfinished metal surfaces of items, components, parts, and assemblies, such as bolts, nuts, washer, saw guard, outrigger guide, and any other exterior metal surfaces requiring a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 or P-19 preservative.



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5.2.1.3.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with type P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2 and secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, as specified for type II, method I.

5.2.1.5 Drive belts and pulleys. Belts shall be removed or released from tension. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.6 Gears.

5.2.1.6.1 Exposed gears. Exposed gears shall be coated with type P-1 or P-19 preservative.

5.2.1.6.2 Enclosed gears. Gear housing shall be filled to the operating level with the operational lubricant and the gears shall be actuated to assure coating of all interior parts and surfaces with the lubricant.

5.2.1.7 Drive chains.

5.2.1.7.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the roller, pins, and bushings. The excess preservative shall be drained and the chain shall be recoated with type P-1 or P-19 preservative. Removed chains shall be coiled, and the coil shall be secured with soft annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to MIL-T-22085 type II. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 or P-19 preservative.

5.2.1.7.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.6.2 for enclosed gears.

5.2.1.8 Depth control.

5.2.1.8.1 Hydraulic type. The MIL-L-21260 fluid (herein referred to as the "hydraulic fluid") supply tank shall be filled to the operating level with approved hydraulic fluid. The pistons shall be fully retracted into the cylinders and secured. Any exposed portions of the piston rods shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wrap shall be secured in place with tape conforming to MIL-T-22085, type II. The hydraulic control valves shall be preserved as specified for piston rods (ramshafts). Hoses shall not be disconnected. Any other metal surface of components of the system requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 or P-19 preservative.



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5.2.1.8.2 Screw-feed type. Exposed threaded portions of the screw feed shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wrap shall be secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.9 Water pump. Interior surfaces of the water pump, valves and piping shall be coated with type P-10, type I, grade 30 preservative oil. Excess oil shall be drained. The shut off cock shall be left open. Other openings shall be sealed with tape conforming to MIL-T-22085, type II or with plastic or metal, caps or plugs conforming to MIL-C-5501.

5.2.1.9.1 Adapters. Each loose adapter shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.10 Saw blades. Each saw blade shall be preserved in accordance with MIL-P-116, method III.

5.2.1.11 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts when they are included.

5.2.1.12 Maintenance tools. Maintenance tools, when included, shall be preserved in accordance with level A requirements of PPP-P-40.

5.2.1.13 Technical publications and manuals. Technical publications and the manual furnished with the engine shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.14 Consolidation. Preserved and unpreserved components and accessories such as saw blades, adapters, saw guide, handles, repair parts, maintenance tools, and technical publications shall be consolidated in the box provided, insofar as space permits, and the box lid secured. Components that cannot be placed in the box provided shall be consolidated in a close-fitting box conforming to PPP-B-636, class weather-resistant, grade as applicable to the weight of the items. Blocking, bracing and cushioning in accordance with MIL-STD-1186 shall be provided to prevent movement and damage to the contents.

5.2.2 Level B. Each complete saw shall be preserved as specified in 5.2.1 for level A with the following exceptions.

5.2.2.1 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be secured to the saw near the pulley for which intended. Pulleys are not required to be coated with primer.

5.2.2.2 Exposed chains. Exposed drive chains shall be preserved as specified in 5.2.1.7.1, except only type P-3 or P-9 preservative shall be applied.

5.2.2.3 Consolidation. Consolidation shall be as specified in 5.2.1.14 except the box shall be class domestic, type CF, variety SW, grade as applicable to the weight of the items to be packaged.



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5.2.3 Level C.

5.2.3.1 Engine. The engine shall be preserved in accordance with the level C requirements of MIL-E-10062.

5.2.3.2 Other components. Components of the saw, other than the engine, shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.3.1 Level A. Each complete saw shall be packed in a close-fitting box conforming to PPP-B-621, class 2, style 7 with hood. The contents shall be blocked, braced, and anchored within the box in accordance with MIL-STD-1186 to prevent free movement or damage to the contents. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class I, type I or IV, finish B size as applicable.

5.3.2 Level B. Each complete saw shall be packed as specified in 5.3.1 for level A except the box shall be class 1 and strapping shall be finish A.

5.3.3 Commercial. Each complete saw preserved as specified in 5.2 shall be individually packed in a container in accordance with ASTM D 3951.

5.4 Depreservative guide. A depreservative guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide", and secured in a conspicuous location on the saw. Unless otherwise specified (see 6.2), DA Form 2258, "Depreservation Guide for Vehicles and Equipment", shall be used.

5.5 Marking.

5.5.1 Military. Packages and shipping containers for levels A and B shall be marked in accordance with MIL-STD-129.

5.5.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, shipping containers shall be marked with the weight and cube data.

6. NOTES

6.1 Intended use. The saw is intended to be used for patching or trenching or for making control joints for portland cement concrete or asphaltic concrete.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When a first article model is not required (see 3.2).
d. Time frame required for submission of first article model (see 3.2).



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e. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When blades shall be furnished and the types, sizes and quantities required (see 3.9).
g. When control identification, safety, warning, or information plates are not to be furnished (see 3.23).
h. Color required when other than as specified (see 3.24).
i. When electromagnetic interference emission conformance is not required (see 3.26).
j. Time intervals for submission of electromagnetic interference reports (see 4.5.2.8).
k. Time frame required for submission of the first article pack (see 5.1).
l. When first article pack test is required (see 4.7.1.2 and 5.1).
m. When DA Form 2258 is not required (see 5.4).
n. Degree of preservation and degree of packing required (see 5.2 and 5.3).
o. Applicable list of repair parts, maintenance tools, and accessories (see 3.28).
p. Type of engine required (see 3.7).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results, and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and contractor's maintenance and operational manual to be furnished with each saw.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools necessary (including any special tools) and instructions on shipment of saw.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the saws in accordance with this specification.

6.7 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use on non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground



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Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research & Development Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Project 3895-0275



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